

Impact of telephone reinforcement of risk reduction education on patient compliance

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Smoking and hyperlipidemia are key risk factors resulting in peripheral arterial disease. These risk factors can be reduced when appropriate lifestyle changes are made, especially for patients less than 60 years of age. Therefore patient education regarding smoking cessation and healthy diet changes may positively impact the course of disease. The ability to make a lifestyle change is affected by (1) an individual's belief that his or her behavior affects the condition and (2) the social support received from family, especially a primary caregiver.

This pilot study compared two randomized groups of patients with peripheral arterial disease for a 1-year period. The control group received the standardized teaching plan regarding smoking cessation and diet at the postoperative clinic visit. The experimental group received the standardized teaching plan in addition to quarterly telephone calls to reinforce smoking cessation and diet information. At the end of 1 year, both groups received follow-up telephone calls at the conclusion of the study for endpoint data collection.

Specifically, this project investigated the following research statements: (1) Patients who receive periodic telephone follow-up will have greater compliance with the recommendations regarding smoking and diet than those who do not receive follow-up. (2) Patients whose caregivers are compliant with the recommendations regarding smoking cessation and diet will have significantly higher compliance themselves.

The data were analyzed by using cross tabulation, sign test, and McNemar's test for nonparametric data to measure mobility, overall perception of health, knowledge of risk reduction information, and compliance with the smoking cessation and low fat/cholesterol diet plan.

The results of this pilot study revealed no significant differences between the control and experimental groups. Telephone reinforcement was not sufficient in this study to influence lifestyle change. However, standardized teaching improved diet scores in both groups. Differences were found between smokers and nonsmokers. (J Vasc Nurs 1998;16:16-20)

It is well known that the implementation of managed care and clinical pathways has significantly shortened patients' length of hospital stays. As a result, when patient teaching is implemented, the patient is more acutely ill, and there is minimal time for reinforcement of new information. During a review of risk reduction education provided to patients with peripheral arterial disease (PAD) at a major midwest teaching hospital, decreased recall and poor integration of information taught during the inpatient session were observed.

Smoking and hyperlipidemia are key risk factors resulting in PAD. These risks can be reduced when appropriate lifestyle changes are made, especially for patients younger than 60 years of age.¹ Some of the lifestyle changes necessary to prevent or slow the progression of PAD include smoking cessation and diet modification. Therefore patient education regarding smoking cessation and diet may have a positive impact on the course of disease if the patient stops smoking, improves diet, or both. The ability to make a lifestyle change is affected by (1) an individual's belief that behavior does in fact affect the condition and (2) the support received from family and friends, especially a primary caregiver.²

Tobacco use is the single most preventable cause of death in the United States, accounting for more than 400,000 deaths per year.^{3,4} The relationship between smoking and the formation of atherosclerosis leading to PAD is widely accepted. Multiple studies have shown that smoking cessation significantly improves survival rate among patients with atherosclerotic cardiovascular disease.^{5,6} Ninety percent of smokers desire to quit, and up to 60% attempt to quit each year. However, only 2% actually succeed at smoking cessation.⁷ Simple office-based education and follow-up can increase this quit rate to 5% to 10% per year.⁸

Dietary factors are associated with half of the leading causes of death in the United States; one of these causes is the development of atherosclerosis.⁹ Atherosclerosis is the underlying process leading to PAD, which begins in childhood and over time results in disease.¹⁰ Chief among the components involved in the diet and PAD relationship is the consumption of foods high in fat, especially saturated fat.¹¹ High dietary fat and cholesterol intake and heredity predisposition lead to a condition known as hyperlipidemia. Extensive evidence exists linking elevated cholesterol levels to the development of atherosclerosis.^{12,13}

This pilot study compared two randomized groups of patients with PAD for a 1-year period. The control group received the standardized teaching plan (Table I) regarding smoking cessation and diet at the postoperative clinic visit. The treatment group received the standardized teaching plan in addition to quarterly telephone calls to reinforce the smoking cessation and diet information. At the conclusion of the study, both groups received follow-up telephone calls to assess their compliance with lifestyle behavior modification.

Specifically, this project investigated the following research hypotheses: (1) Patients who receive periodic telephone follow-up will have greater compliance with the recommendations regarding smoking and diet than those who do not receive follow-up. (2) Patients whose caregivers are compliant with the recommendations regarding smoking cessation and diet will have significantly higher compliance themselves.

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REVIEW OF THE LITERATURE

According to the American Heart Association, cardiovascular disease continues to be the leading cause of death in the United States. On the basis of these data gathered by the Association, it is estimated that 1 in 4 Americans have at least one form of cardiovascular disease. PAD is a chronic and disabling illness that primarily affects the middle-aged to elderly population.¹⁴ Mounting research has shown the correlation between one's lifestyle behavior and the formation of atherosclerosis.^{15,16} Modifiable and nonmodifiable risk factors are associated with the development of atherosclerosis. Modifiable factors include smoking, hyperlipidemia, diabetes mellitus, hypertension, sedentary lifestyle, and stress. Nonmodifiable factors are age, gender, and family history.¹⁷

The single most significant factor contributing to both the onset and progression of PAD is smoking.¹⁴ This fact was strikingly demonstrated in a study conducted by the Mayo Clinic in which nearly half the subjects diagnosed with atherosclerotic PAD abstained from smoking for 5 years. Of the patients who continued to smoke, 11% required limb amputation within this time period.¹⁸

Elevated total serum cholesterol, low-density lipoproteins, and triglycerides are associated with the development of PAD.¹⁹ The National Cholesterol Education Commission recommends that men and women of all ages older than 20 years reduce total serum cholesterol levels to less than 200 mg/dL.²⁰

Noncompliance is a significant health care problem today. It has been noted that more than half the patients for whom appropriate therapy is prescribed fail to receive full benefit through inadequate adherence to treatment.²¹ A positive correlation exists between compliance and social support. Social support has been shown to be directly related to behavior change that can lead to healthy lifestyles.²² Therefore it is critical to examine social support when measuring compliance of risk reduction counseling.

Today's health care environment emphasizes primary and preventive care through increased screening efforts and patient education. Patient education is the primary intervention used by nurses to promote self-care for their patients. Previous research has shown that recall of patient teaching in the hospital is often poor.²³ The physical environment of the hospital unit is often not private, quiet, or comfortable. The patient may be experiencing physical discomfort or the effects of medications, which may hinder learning. In addition, patients and caregivers have a high level of anxiety when hospitalized.¹⁶ The current health care environment dictates that nurses' roles will continue beyond the time of discharge. Patient education is an ongoing process that needs periodic reinforcement.²⁴ Nurses are encouraged to reinforce newly learned information through patient follow-up in the outpatient arena and by telephone follow-up.²⁵⁻²⁷

PROCEDURE

This was a pretest-posttest control group design. The sample included male and female patients younger than 60 years of age with atherosclerotic PAD. This age group was selected because it is believed that (1) lifestyle modification,

TABLE I

COMPLETE LIFESTYLE BEHAVIOR QUESTIONNAIRE WITH PATIENT

Diet education

Definition of cholesterol, high-density lipoproteins, and low-density lipoproteins

Fat/cholesterol and atherosclerosis

Dietary cholesterol

Differences between saturated, monounsaturated, and polyunsaturated fats

Interpretation of food labels

Calculation of fat in diet

Recipe suggestions

Smoking cessation education

Smoking and atherosclerosis

Benefits to smoking cessation

Cessation strategies

Methods to smoking cessation

Establish *mutually* acceptable goals for lifestyle behavior change

such as smoking cessation and diet, will have a greater impact on early disease; and (2) some studies have shown negative benefits of cholesterol reduction for the elderly.¹ The following criteria were met for entrance into the study:

1. Admitted by a vascular surgery attending physician with a diagnosis of PAD
2. Speak and read English
3. Have one or both of the following modifiable risk factors: smoking and hyperlipidemia
4. Able to give informed consent

Participants were randomly assigned to the control or experimental group after the 2 to 3 week postdischarge outpatient clinic visit. See Figure 1 to schematically view the overall research design. This study was implemented as part of the annual quality evaluation program in the vascular center.

PATIENT EDUCATION

The patient's current lifestyle behavior was evaluated by a questionnaire derived from the Mos-36-item short-form health survey (SF-36).²⁸ The final questionnaire used in this study assessed mobility, health perception, knowledge of PAD or risk factors for PAD, and current smoking and diet practices. The structure of the interview included preparing the environment, developing a therapeutic relationship, identifying the problem, educating, and mutual goal setting.²⁹ Table I summarizes the education plan. All of the information discussed during the teaching session was adapted by the researchers from current literature and supported in writing by patient educational material from the American Heart Association and The Society for Vascular Nursing.

TABLE II**FINDINGS****Overall**

- No significant difference between control and experimental groups
- Caregiver component was eliminated because of lack of caregiver presence
- Decreased retention of learned information as time progressed

Demographics

21 total study participants

	Control group (N = 10)	Experimental group (N = 11)
Average age	49 yr	53 yr
Gender		
Male	80%	72.7%
Female	20%	27.3%
Smoker		
Yes	90%	81.8%
No	10%	18.2%

Lifestyle change

- Telephone reinforcement was not sufficient to influence overall lifestyle change
- Physician reinforcement and support appeared to influence the degree of lifestyle change
- A trend revealed an improvement in mobility and health perception in both groups as time progressed
- Patients who had higher perceived mobility were more likely to meet goals

Diet

- Standardized teaching improved diet scores in both groups

Smoking

- Important differences between smokers and nonsmokers were noted:
 - Nonsmokers had more improved diet scores and were more compliant in meeting goals
 - Smokers had a higher perception of health at completion of study
- Only one patient in study quit smoking
- No smokers agreed to attend a formal smoking cessation class

SETTING

This study took place in the vascular center of a major teaching hospital. The outpatient clinic had four patient consultation rooms that were conducive to learning and teaching. The rooms were private, well lit, and of a comfortable temperature. The telephone follow-up phone calls were made by two advanced practice nurses within the vascular center.

DATA COLLECTION

Twenty-one participants were enrolled in this pilot study. See Table II for the demographics. Data collection occurred at the initial interview for both the control and experimental groups. The experimental group received quarterly telephone calls to assess change in mobility, health perception, retention of learned information, compliance toward goals, and reinforcement of smoking cessation and diet information. One participant was eliminated from the study because of the inability to reach the client at a data collection point.

The control group only received one telephone call at the end of 1 year for comparison data collection.

DATA ANALYSIS

The data were analyzed by cross tabulation, sign test, and McNemar's test for nonparametric data to measure mobility, overall perception of health, knowledge of risk reduction information, and compliance with the smoking cessation and low fat-low cholesterol diet plan.

FINDINGS

The results of this pilot study revealed no significant differences between the control and experimental groups. Telephone reinforcement was not sufficient in this study to influence lifestyle change. However, standardized teaching improved diet scores in both groups. The caregiver component was eliminated because of lack of caregiver presence. Differences were found between smokers and nonsmokers.

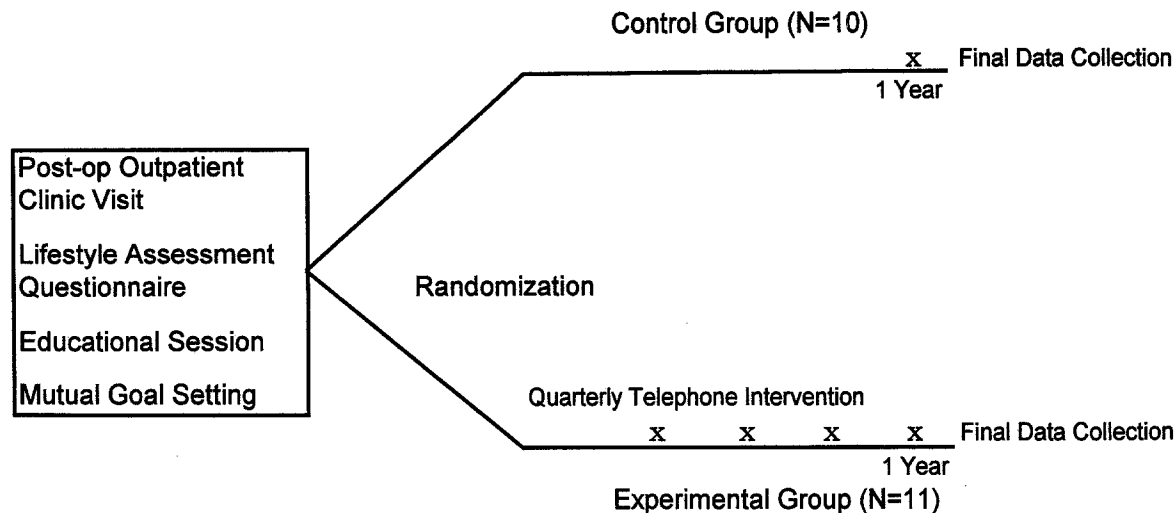


Figure 1. Overall research design.

Nonsmoking individuals had more improved diet scores and were more compliant as measured by goals met. Interestingly, smokers had a higher perception of health at the completion of the study. Only one person during the study quit smoking. None of the patients in the study agreed to attend a formal smoking cessation class. One physician who consistently reinforced the importance of lifestyle change had proportionately more patients who met their goals. This reflects the importance of nurse and physician collaboration and educational reinforcement. It is important to note that patients who met their goals had higher perceived mobility. Finally, the endpoint telephone call for the control group revealed decreased retention of learned information as time progressed. This group received no periodic reinforcement of learned information. See Table II for summarized findings.

NURSING IMPLICATIONS

Patient education is an essential independent component of nursing practice. To validate the impact of patient education-related interventions, outcome focused studies such as this need to be performed. Recommendations for future research include a larger sample size, more frequent phone calls, and conducting the initial lifestyle assessment at the time of diagnosis, not after surgery. Researchers of this study observed that many lifestyle changes are made immediately at the time of diagnosis. Gathering initial baseline information after instead of before surgery limited the degree of behavior change observed in the final data analysis. The reduced length of stay experienced by patients today threatens the efficacy of patient learning. Follow-up telephone sessions continue to offer hope for bridging the gap between information learned in the hospital and lifestyle behavior change at home.

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